

Variational Methods Exam Preparation

Part 2 recap: variational approximation and finite elements

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1 How To Use This Sheet

This document summarizes what to learn from the Part 2 lecture slides and the matching worksheets.

Primary sources:

- Lectures/VMAM_slides_part2.pdf: main Part 2 lecture slides.
- Lectures/td5-1.pdf: unisolvent finite element worksheet.
- Lectures/td6.pdf: variational approximation and finite element worksheet.
- Lectures/poly_MVAM.pdf: global lecture notes, especially Chapter 4.

Priority: learn the theoretical approximation framework first, then learn how it becomes a finite element matrix, then practice unisolvency and small mesh computations.

2 Inner Variational Approximation

Sources: VMAM_slides_part2.pdf, slides 3-11. Context: poly_MVAM.pdf, Chapter 4.1, pages 21-22.

The continuous variational problem is

$$(FV) \quad \text{Find } u \in V \text{ such that } a(u, v) = \ell(v) \quad \forall v \in V.$$

The approximate problem is

$$(FV_h) \quad \text{Find } u_h \in V_h \text{ such that } a(u_h, v_h) = \ell(v_h) \quad \forall v_h \in V_h.$$

Here V_h is finite-dimensional and $V_h \subset V$. This is called an **inner approximation** because the discrete space stays inside the continuous energy space.

What you must know:

- Why $V_h \subset V$ matters.
- Why V_h is a Hilbert space: it is finite-dimensional and closed in V .
- Why Lax-Milgram applies on V_h with the same continuity and coercivity constants as on V .
- Why the discrete solution u_h is unique.

If $(\varphi_1, \dots, \varphi_N)$ is a basis of V_h , then

$$u_h = \sum_{j=1}^N u_h^j \varphi_j.$$

Testing only with the basis functions gives the linear system

$$\sum_{j=1}^N a(\varphi_j, \varphi_i) u_h^j = \ell(\varphi_i), \quad i = 1, \dots, N.$$

Equivalently,

$$AU = L, \quad A_{ij} = a(\varphi_j, \varphi_i), \quad L_i = \ell(\varphi_i).$$

Important detail: the order $A_{ij} = a(\varphi_j, \varphi_i)$ follows from testing by φ_i and expanding u_h over φ_j .

If a is coercive, then A is positive definite and therefore invertible.

3 Cea's Lemma

Sources: VMAM_slides_part2.pdf, slides 8-10. Context: poly_MVAM.pdf, Theorem 4.1, pages 21-22.

This is the central theorem of Part 2.

$$\|u - u_h\|_V \leq C \inf_{v_h \in V_h} \|u - v_h\|_V.$$

Interpretation: the finite element error is bounded by the best approximation error in the chosen finite-dimensional space.

The proof idea:

- Since $V_h \subset V$, every $w_h \in V_h$ is an admissible test function in the continuous problem.
- Subtract the two variational formulations.
- Obtain Galerkin orthogonality:

$$a(u - u_h, w_h) = 0 \quad \forall w_h \in V_h.$$

- For any $v_h \in V_h$, use $w_h = v_h - u_h$.
- Then

$$a(u - u_h, u - u_h) = a(u - u_h, u - v_h).$$

- Use coercivity and continuity:

$$\alpha \|u - u_h\|_V^2 \leq M \|u - u_h\|_V \|u - v_h\|_V.$$

- Divide by $\|u - u_h\|_V$ and take the infimum over $v_h \in V_h$.

What to remember for the exam: Cea's lemma turns convergence into a pure approximation question.

4 Meaning Of $V_h \rightarrow V$

Sources: VMAM_slides_part2.pdf, slide 10. Context: poly_MVAM.pdf, Theorem 4.2, page 22.

The lecture makes precise what $V_h \rightarrow V$ means.

Assume there is a dense subspace $\mathcal{V} \subset V$ and operators

$$r_h : \mathcal{V} \rightarrow V_h$$

such that

$$\forall v \in \mathcal{V}, \quad \|v - r_h v\|_V \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

Then the discrete solution converges:

$$\|u - u_h\|_V \rightarrow 0.$$

Interpretation:

- r_h is usually an interpolation or projection operator.
- Density lets you approximate any $u \in V$ by a nicer function $u_\varepsilon \in \mathcal{V}$.
- The operator r_h then approximates u_ε by an element of V_h .
- Cea's lemma finishes the convergence proof.

5 Hilbert Basis Galerkin Approximation

Sources: VMAM_slides_part2.pdf, slides 12-17. Context: poly_MVAM.pdf, Chapter 4.2, pages 22-25.

A Hilbert basis of V is an orthonormal countable family $(e_n)_{n \geq 1}$ whose linear span is dense in V .

One can choose

$$V_h = \text{span}(e_1, e_2, \dots, e_m), \quad h = \frac{1}{m}.$$

Then the Galerkin solution u_m converges to u .

The important message is not only convergence. The lecture emphasizes that this approach is mostly theoretical.

Advantages:

- It proves existence by reducing infinite-dimensional problems to finite-dimensional linear systems.
- It gives a clean conceptual version of Galerkin approximation.

Disadvantages:

- Matrices are often full.
- Conditioning can be very bad.
- Boundary conditions can be difficult to encode for non-academic domains.

The monomial example in slides 14-16 is important. It converges fast for small m , but the condition number is already around 10^8 for $m = 7$. Lesson: good approximation is not enough; the matrix must also be numerically usable.

6 Finite Element Meshes

Sources: VMAM_slides_part2.pdf, slides 18-19. Context: poly_MVAM.pdf, Chapter 4.5.1, page 32.

Finite element methods choose V_h using localized basis functions built from a mesh of the domain.

A conforming mesh satisfies:

- The union of all elements covers Ω .
- Element interiors do not overlap.
- Elements are not flat.
- An internal edge is always the full edge of the neighboring element.

For a triangle T_l , define the diameter

$$h_{T_l} = \max_{(x_1, x_2) \in T_l^2} \|x_1 - x_2\|.$$

Define the roundness by the diameter of the largest inscribed ball:

$$\rho_{T_l} = \max_{B_r \subset T_l} 2r.$$

The ratio

$$\frac{h_{T_l}}{\rho_{T_l}} \geq 1$$

measures flatness. Good meshes avoid large values of this ratio.

Why finite elements give sparse matrices: each basis function is supported only on a small patch of neighboring elements, so most products $a(\varphi_j, \varphi_i)$ vanish.

7 P1 Lagrange Finite Elements

Sources: VMAM_slides_part2.pdf, slides 21-25. Context: poly_MVAM.pdf, Chapter 4.5.2, pages 32-33.

The P_1 Lagrange finite element space is

$$V_h = \{v_h \in C^0(\overline{\Omega}) : v_h|_{T_l} \in P_1, \quad \forall l = 1, \dots, N_t\}.$$

In two dimensions,

$$P_1 = \text{span}(1, x, y), \quad \dim P_1 = 3.$$

Key facts:

- $V_h \subset H^1(\Omega)$ because functions are continuous and piecewise H^1 .
- A P_1 finite element function is completely determined by its values at mesh vertices.
- If the mesh has N_v vertices, then $\dim V_h = N_v$.
- The global basis functions are the hat functions φ_i satisfying

$$\varphi_{i(a_j)} = \delta_{ij}.$$

Every $v_h \in V_h$ can be written as

$$v_h(x) = \sum_{i=1}^{N_v} v_h(a_i) \varphi_{i(x)}.$$

For homogeneous Dirichlet boundary conditions, use

$$\tilde{V}_h = V_h \cap H_0^1(\Omega).$$

Equivalently,

$$\tilde{V}_h = \{v_h \in V_h : v_h(a_i) = 0 \text{ for all boundary vertices } a_i\}.$$

Thus only the basis functions associated with interior vertices remain:

$$\tilde{V}_h = \text{span} \{ \varphi_i : a_i \notin \partial\Omega \}.$$

8 Finite Element Triples And Unisolvency

Sources: VMAM_slides_part2.pdf, slides 26-28. Context: poly_MVAM.pdf, Chapter 4.4, pages 30-31.

Worksheet: td5-1.pdf.

A finite element is a triple

$$(K, P, \Sigma),$$

where:

- K is a non-empty compact subset of $\mathbb{R}^d$.
- P is a finite-dimensional vector space of functions on K .
- $\Sigma = \{\ell_1, \ell_2, \dots, \ell_m\}$ is a set of linear forms on P .

The numbers $\ell_{j(p)}$ are the local degrees of freedom.

The finite element is **unisolvent** if

$$\forall (\alpha_1, \dots, \alpha_m) \in \mathbb{R}^m, \exists! p \in P \text{ such that } \ell_{j(p)} = \alpha_j \text{ for } j = 1, \dots, m.$$

Exam method for proving unisolvency:

- First check $\text{card}(\Sigma) = \dim P$.
- If the dimensions do not match, the element is not unisolvent.
- If the dimensions match, prove that the only $p \in P$ such that $\ell_{j(p)} = 0$ for all j is $p = 0$.
- Equivalently, construct local basis functions φ_i such that

$$\ell_{j(\varphi_i)} = \delta_{ij}.$$

Worksheet td5-1.pdf trains exactly this skill.

What to practice there:

- Exercise 1: square with vertex values, center value, and one integral degree of freedom.
- Exercise 2: triangle edge nodes and barycentric coordinates.
- Exercise 3: polynomial space involving $\lambda_1, \lambda_2, \lambda_3$, and values at vertices, midpoints, and barycenter.

9 Barycentric Coordinates

Sources: poly_MVAM.pdf, Chapter 4.5.2, pages 32-33. Worksheet: td5-1.pdf, Exercises 2 and 3.

For a triangle with vertices a_1, a_2, a_3 , the barycentric coordinates of a point x are $(\lambda_1, \lambda_2, \lambda_3)$ such that

$$x = \lambda_1 a_1 + \lambda_2 a_2 + \lambda_3 a_3, \quad \lambda_1 + \lambda_2 + \lambda_3 = 1.$$

Important values:

$$a_1 : (1, 0, 0), \quad a_2 : (0, 1, 0), \quad a_3 : (0, 0, 1).$$

At the midpoint a_{ij} of edge $[a_i, a_j]$,

$$\lambda_i = \lambda_j = \frac{1}{2}, \quad \lambda_k = 0.$$

For P_1 interpolation,

$$p = \sum_{i=1}^3 p(a_i) \lambda_i.$$

For P_2 interpolation,

$$p = \sum_{i=1}^3 p(a_i) \lambda_i (2\lambda_i - 1) + 4 \sum_{1 \leq i < j \leq 3} p(a_{ij}) \lambda_i \lambda_j.$$

This formula is especially useful for constructing basis functions in td5-1.pdf.

10 Affine Equivalence

Sources: VMAM_slides_part2.pdf, slides 29-31. Context: poly_MVAM.pdf, Proposition 4.2, page 31.

Finite element calculations are usually performed on a reference element and transported to each physical element.

For triangles, the affine map has the form

$$F(\hat{x}) = B\hat{x} + b.$$

If $(\hat{K}, \hat{P}, \hat{\Sigma})$ is the reference element and (K, P, Σ) is the physical element, the local basis functions transform by

$$\varphi_i = \hat{\varphi}_i \circ F^{-1}.$$

Practical meaning: instead of deriving basis functions and integrals separately on every triangle, compute them on the reference triangle and map them by F .

11 Higher-Order Elements

Sources: VMAM_slides_part2.pdf, slides 32-35.

The P_k Lagrange finite element space is

$$V_h^k = \{v_h \in C^0(\bar{\Omega}) : v_h|_{T_l} \in P_k, \forall l\}.$$

Increasing k has three effects:

- The matrix size increases.
- The matrix stays sparse, but has more non-zero terms.
- The approximation is better if the exact solution is regular enough.

The standard estimate is

$$\|u - u_h\|_{H^1(\Omega)} \leq Ch^k \|u\|_{H^{k+1}(\Omega)}.$$

Examples to recognize:

- P_2 in 1D uses nodes 0, $\frac{1}{2}$, and 1.
- P_2 in 2D uses vertices and edge midpoints.
- P_3 in 2D adds more edge nodes and one interior node.
- Q_1 on a square uses $P = \text{span}(1, x, y, xy)$.
- Hermite elements use function values and derivative values as degrees of freedom, and are useful for fourth-order problems.

12 One-Dimensional P1 Assembly

Sources: poly_MVAM.pdf, Chapter 4.3.2, pages 25-28. Worksheet: td6.pdf, Exercise 1.

For the model problem

$$-u'' + u = f \quad \text{on } (0, 1), \quad u(0) = u(1) = 0,$$

the variational formulation is:

$$\text{Find } u \in H_0^1(0, 1) \text{ such that } \int_0^1 u'v' \, dx + \int_0^1 uv \, dx = \int_0^1 fv \, dx \quad \forall v \in H_0^1(0, 1).$$

For uniform P_1 elements, the basis functions are the hat functions φ_i .

The bilinear form is

$$a(u, v) = \int_0^1 u' v' \, dx + \int_0^1 uv \, dx.$$

The standard coefficients are

$$a(\varphi_i, \varphi_i) = \frac{2}{h} + 2\frac{h}{3},$$

and

$$a(\varphi_i, \varphi_{i+1}) = -\frac{1}{h} + \frac{h}{6}.$$

The matrix is tridiagonal because φ_i and φ_j have disjoint supports when $|i - j| > 1$.

In `td6.pdf`, Exercise 1, you should be able to:

- Write the variational formulation.
- Identify the Hilbert space V .
- Define the discrete space V_h .
- Define the hat basis functions.
- Write the discrete variational problem.
- Write the final linear system with explicit coefficients.
- Explain how to treat a non-homogeneous Dirichlet condition $u(0) = b$.

13 Two-Dimensional P1 Computations On Small Meshes

Sources: `td6.pdf`, Exercises 2 and 3. Context: `VMAM_slides_part2.pdf`, slides 21-25, and `poly_MVAM.pdf`, Chapter 4.5.

For homogeneous Dirichlet boundary conditions, only interior vertices give degrees of freedom.

In `td6.pdf`, Exercise 2, the square is divided into four triangles and there is one interior point E . Therefore the discrete space has dimension 1.

Let w be the unique basis function satisfying

$$w(E) = 1, \quad w = 0 \quad \text{at all boundary vertices.}$$

Then

$$u_h = Uw.$$

The linear system is a single equation

$$aU = b,$$

where

$$a = \int_{\Omega} |\nabla w|^2 \, dx \, dy, \quad b = \int_{\Omega} w \, dx \, dy.$$

The worksheet result is

$$U = \frac{1}{12}.$$

For `td6.pdf`, Exercise 3, the mesh is a long rectangle subdivided into repeated triangular cells. The expected skills are:

- Count the interior degrees of freedom.
- Define the P_1 approximation space.

- Give the hat-function basis explicitly by nodes.
- Use symmetry/repetition of the mesh to solve the discrete problem efficiently.

14 Exam Priority Checklist

1. Understand the passage from (FV) to (FV_h) .
2. Derive the matrix system $AU = L$ from a basis of V_h .
3. Prove Cea's lemma.
4. Explain the convergence criterion using $r_h : \mathcal{V} \rightarrow V_h$.
5. Define P_1 finite elements on a triangulation.
6. Prove $\dim V_h = N_v$ for P_1 elements.
7. Explain how homogeneous Dirichlet conditions remove boundary basis functions.
8. Define a finite element triple (K, P, Σ) .
9. Test unisolvency by dimension count and zero-kernel argument.
10. Construct local basis functions with barycentric coordinates.
11. Explain affine equivalence and reference elements.
12. Compute simple 1D P_1 stiffness and mass entries.
13. Compute simple 2D P_1 examples on tiny meshes.

15 Likely Exam Questions

- State and prove Cea's lemma.
- Explain why the discrete variational problem has a unique solution.
- Starting from a basis of V_h , derive the matrix system.
- Define the P_1 finite element space and prove that its dimension is the number of mesh vertices.
- Explain why $V_h \subset H^1(\Omega)$.
- Explain how homogeneous Dirichlet boundary conditions are imposed in finite elements.
- Decide whether a finite element is unisolvent.
- Build basis functions using barycentric coordinates.
- Compute a small stiffness matrix by hand.
- Explain why finite element matrices are sparse.
- Compare Hilbert-basis Galerkin methods with finite element methods.

16 Minimal Proof Templates To Memorize

16.1 Cea's Lemma Template

Start from Galerkin orthogonality:

$$a(u - u_h, w_h) = 0 \quad \forall w_h \in V_h.$$

For arbitrary $v_h \in V_h$, choose $w_h = v_h - u_h$. Then

$$a(u - u_h, u - u_h) = a(u - u_h, u - v_h).$$

By coercivity and continuity,

$$\alpha \|u - u_h\|_V^2 \leq M \|u - u_h\|_V \|u - v_h\|_V.$$

Conclude

$$\|u - u_h\|_V \leq \left(\frac{M}{\alpha} \right) \|u - v_h\|_V.$$

Then take the infimum over $v_h \in V_h$.

16.2 Unisolvency Template

To prove that (K, P, Σ) is unisolvent:

1. Check $\text{card}(\Sigma) = \dim P$.
2. Let $p \in P$ satisfy $\ell(p) = 0$ for every $\ell \in \Sigma$.
3. Use the conditions to force all coefficients of p to vanish.
4. Conclude $p = 0$.
5. Therefore the map $p \rightarrow (\ell_1(p), \dots, \ell_{m(p)}(p))$ is injective between spaces of the same finite dimension, hence bijective.

To prove it is not unisolvent, find one non-zero $p \in P$ such that all degrees of freedom vanish.

16.3 Homogeneous Dirichlet Finite Element Template

Start from

$$V_h = \{v_h \in C^0(\overline{\Omega}) : v_h|_T \in P_1\}.$$

For homogeneous Dirichlet conditions, use

$$V_{h,0} = V_h \cap H_0^1(\Omega).$$

Since P_1 functions are determined by vertex values, impose

$$v_{h(a_i)} = 0$$

for every boundary vertex a_i . The remaining unknowns are the values at interior vertices.